JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CHEMISTRY 8872/01

Paper 1 Multiple Choice

20 September 2011

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class and shade your exam index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record you choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

SECTION A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1. Carbon disulfide vapour burns in oxygen according to the following equation.

$$CS_2(g) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$$

A sample of 10 cm^3 of carbon disulfide was burnt in 50 cm^3 of oxygen. After measuring the volume of gas remaining, the product was treated with an excess of limewater and the volume of gas measured again. All measurements were made at the same temperature and pressure in which carbon disulfide was gaseous.

What were the measured volumes?

	volume of gas after burning / cm^3	volume of gas after treated with limewater/ cm^3
A	30	0
B	30	20
C	50	20
D	50	40

2. 10 cm^3 of 0.10 mol dm^{-3} aqueous iron(II) sulfate is titrated against 0.025 mol dm^{-3} aqueous potassium manganate(VII) in the presence of excess hydrogen ions. It is found that exactly 10 cm^3 of the manganate(VII) solution is required to reach the endpoint.

What is the oxidation number of the manganese at the endpoint?

A +5 **B** +4 **C** +3 **D** +2

3. Which of the following ions has more electrons than protons and more protons than neutrons?

[H = ${}^{1}_{1}H$; D = ${}^{2}_{1}H$; He = ${}^{4}_{2}He$; O = ${}^{16}_{8}O$]

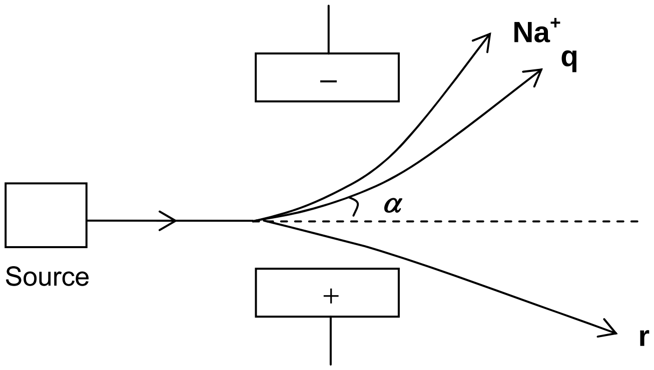
A OH^-

B He^+

C D_3O^+

D OD^-

4 In an experiment, a sample was vaporised, ionised and passed through an electric field. Analysis of the deflection occurring at the electric region revealed the following data for the sample. It was observed that a beam of Na^+ gives an angle of deflection of 4°.



What are the possible identities of unknown particles, **q** and **r**, and the value of α ?

	q	**r**	α
A	Be^{2+}	S^{2-}	2.0
B	Ba^{2+}	S^{2-}	1.3
C	Ba^{2+}	Br^-	2.0
D	Be^{2+}	Br^-	1.3

5. Which of the following elements would be expected to form the largest ion with a noble gas electron configuration?

A Aluminum

B Chlorine

C Phosphorus

D Potassium

6. BH_3 reacts with $(CH_3)_3N$ to give $(CH_3)_3NBH_3$. Which one of the following will be true of $(CH_3)_3NBH_3$?

A The N atom is electron rich.

B It contains hydrogen bonding.

C The B atom is electron deficient.

D The B atom is tetrahedrally coordinated.

7. Which one of the compounds is ionic and contains both sigma and pi bonds?

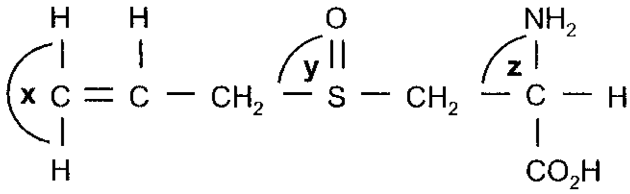
A $Fe(OH)_3$

B HC*l*O

C H_2S

D NaCN

8. The characteristic smell of garlic is due to alliin. When garlic is cut, an enzyme releases alliin because alliin has the right shape to fit the enzyme molecule.



alliin

What would you expect the bond angles x, **y** and **z** to be (to the nearest 10°) in a molecule of alliin?

	x	y	z
A	120°	120°	90°
B	120°	107°	110°
C	180°	107°	90°
D	180°	110°	110°

9. Silicon carbide (carborundum) is a shiny, hard, chemically inert material with a very high melting point. It can be used to sharpen knives and make crucibles.

Which type of structure explains these properties?

A a giant layer structure with covalent bonds between atoms and van der Waals' forces between the layers

B a simple molecular structure with covalent bonds between the atoms of silicon and carbon

C a giant structure with covalent bonds between silicon and carbon atoms

D a giant structure containing ionic bond between Si^{4+} and C^{4-}.

10. Which of the following does the enthalpy change ΔH associated with a chemical change **not** depend upon?

A The amount of the reactants.

B The physical states of the reactants.

C The pressure at which the change occurs.

D The number of stages involved in the change.

11. Consider the data in the table below.

substance	Standard enthalpy change (heat) of combustion /kJ mol^{-1}
hydrogen (g)	–300
carbon (s)	–400
benzene (*l*)	–3350

What is the standard enthalpy change of formation of liquid benzene?

A –4050 kJ mol^{-1} **B** +2650 kJ mol^{-1} **C** –50 kJ mol^{-1} **D** +50 kJ mol^{-1}

12. Which ionic compound has the most exothermic lattice energy?

A sodium chloride

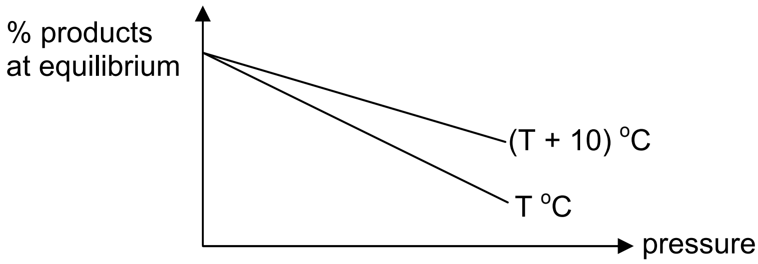
B sodium fluoride

C lithium fluoride

D lithium iodide

13. The graphs below show how the percentage of gaseous products present at equilibrium vary with temperature and pressure.

Which of the following systems could the graph represent?



A $H_2 (g) + I_2 (g) \rightleftharpoons 2HI (g)$ $\Delta H = +53$ kJ mol^{-1}

B $N_2 (g) + 3H_2 (g) \rightleftharpoons 2NH_3 (g)$ $\Delta H = -92$ kJ mol^{-1}

C $N_2O_4 (g) \rightleftharpoons 2NO_2 (g)$ $\Delta H = +57$ kJ mol^{-1}

D $2Fe (s) + \frac{3}{2} O_2 (g) \rightleftharpoons Fe_2O_3 (s)$ $\Delta H = -822$ kJ mol^{-1}

14. Iodine–131 is a radioactive isotope with a half–life of 8 days. Following the nuclear power plant disaster at Chernobyl in 1986, it was stated that the situation would become safe only when $\frac{1}{1024}$ of the iodine–131 isotopes remained in the cloud of vapour formed.

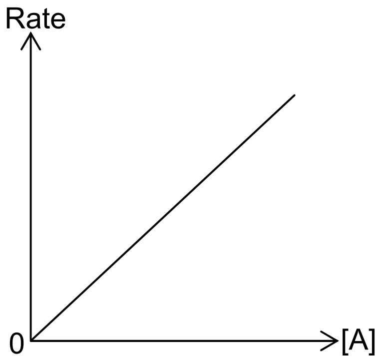
Given that the radioactive decay is a first–order reaction, how long would it take for the situation to become safe?

A 1024 days **B** 80 days **C** 72 days **D** 40 days

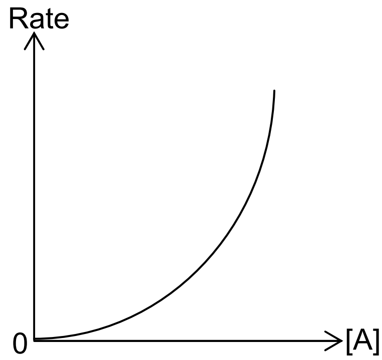
15. Consider the hypothetical reaction: $2A \rightarrow B + C$.

Given that the rate constant, k, of the reaction is 0.156 s^{-1} , which of the following graphs correctly reflects the reaction kinetics of the reaction?

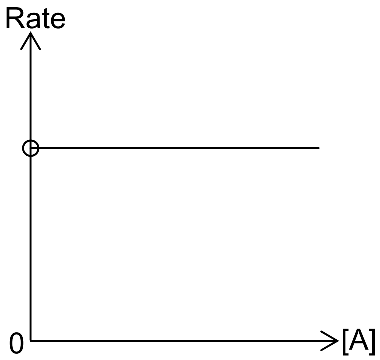
A



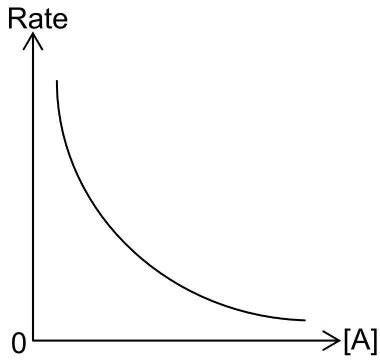
C



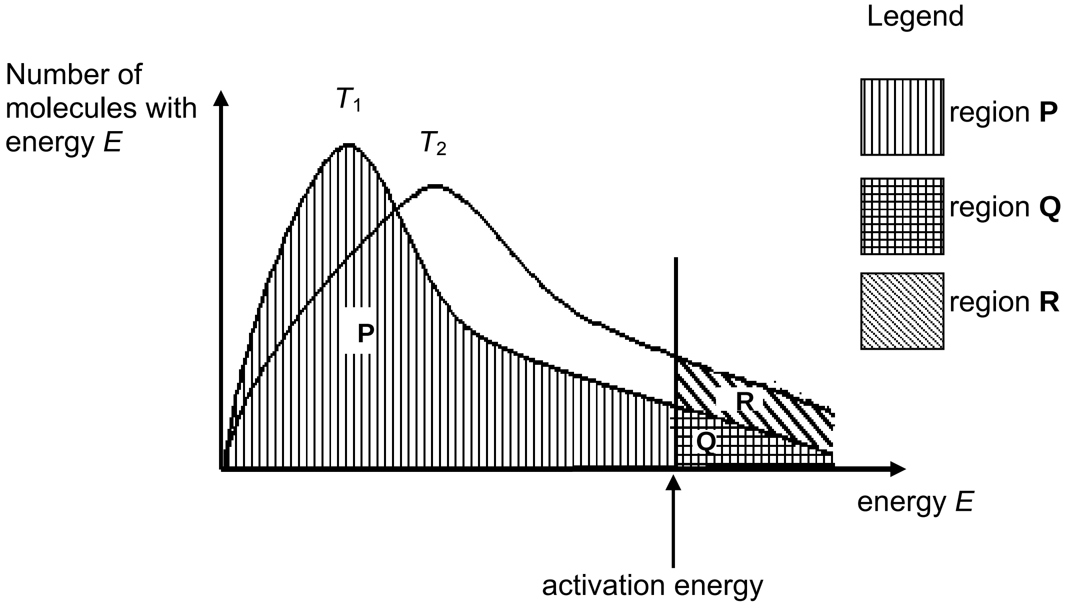
B



D



16. The distribution of the number of molecules with energy *E* is given in the sketch for two temperatures, T_1 and a higher temperature T_2. The letters *P*, *Q*, *R* refer to the separate and differently shaded areas. The activation energy is marked on the energy axis.



Which of the following statements is correct when temperature is increased from T_1 to T_2?

A The fraction of the molecules which have energy $\geq$ activation energy at T_2 is represented by $\dfrac{Q+R}{P+Q}$.

B The frequency of collisions between molecules is greater at a higher temperature.

C The proportion of molecules with energies above any given value decreases.

D The proportion of molecules with any given energy increases.

17. **P**, **Q** and **R** are main group elements in the same period of the Periodic Table. The oxide of **P** is basic, the oxide of **Q** is acidic and the oxide of **R** is amphoteric.

What is the order of increasing atomic number for these elements, starting from the lowest?

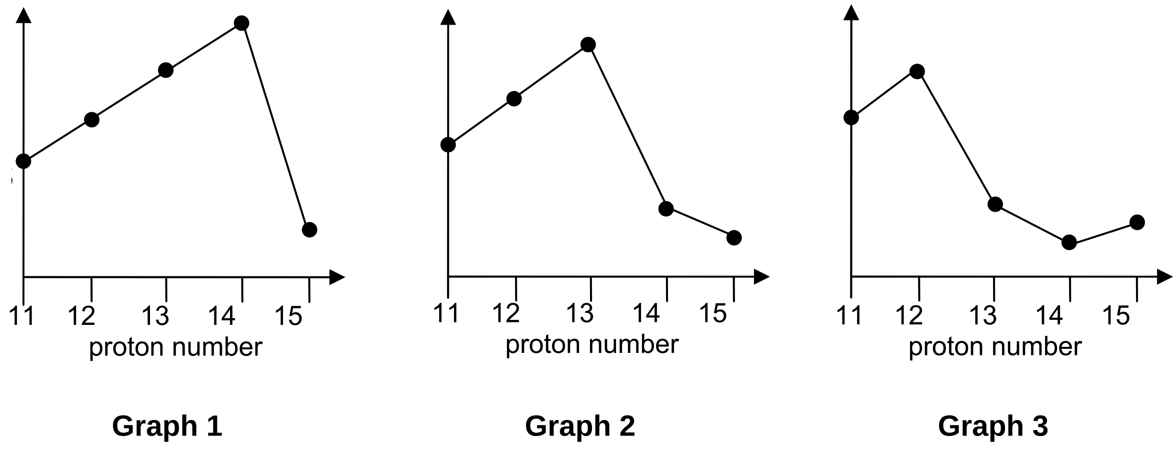
A **P** < **Q** < **R**

B **P** < **R** < **Q**

C **Q** < **R** < **P**

D **R** < **P** < **Q**

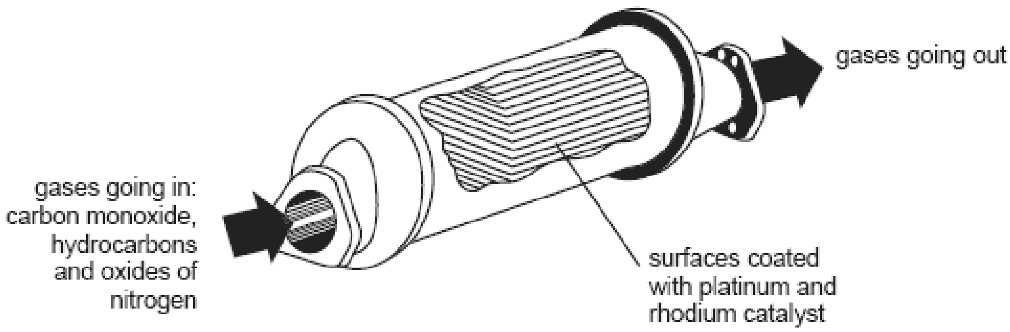
18. The following graphs show how three properties of the elements, Na to P, and their compounds, vary with proton number.



What properties are shown by the three graphs?

	Graph 1	Graph 2	Graph 3
A	Melting point of oxide	Melting point of chloride	Conductivity of element
B	Melting point of chloride	Melting point of element	Melting point of oxide
C	Melting point of oxide	Conductivity of element	Melting point of chloride
D	Melting point of element	Melting point of chloride	Conductivity of element

19. Which reaction in the catalytic converter does **not** remove hazardous and polluting gases from the exhaust fumes of a motor car?



These equations are qualitative and unbalanced.

[HC = unburnt hydrocarbon; NO_x = oxides of nitrogen]

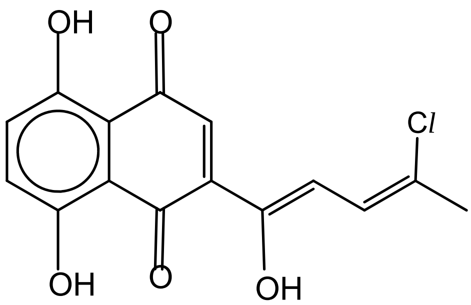
A $HC + NO_x \longrightarrow H_2O + CO + N_2$

B $CO + NO_x \longrightarrow CO_2 + N_2$

C $HC + NO_x \longrightarrow H_2O + CO_2 + N_2$

D $CO + O_2 \longrightarrow CO_2$

20. How many geometric isomers are there in the compound shown?



A 2 **B** 3 **C** 4 **D** 8

21. Ethane reacts with chlorine gas in the presence of ultraviolet light to form a mixture of products via free radical substitution. Which statement is true about this reaction?

A Chloroethane is formed only in the propagation step.

B Bond formation occurs only in the termination step.

C Homolytic fission occurs only in the initiation step.

D Butane is formed only in the termination step.

22. Aldehydes and ketones are produced industrially by the catalytic oxidation of alkenes, e.g. ethanal is manufactured from ethene as shown below.

$$H_2C{=}CH_2 + \tfrac{1}{2}O_2 \xrightarrow{\text{catalyst}} CH_3CHO$$

This process is also used industrially with but-2-ene.

Which of the following represents the structure of the compound which would be produced from but-2-ene?

A $CH_3CH_2CH_2CHO$

B $CH_3COCH_2CH_3$

C $CH_3CO(CH_3)_2$

D $(CH_3)_2CHCHO$

23. Some alcohols were separately treated with hot concentrated H_2SO_4. Two of these gave the same molecular formula, C_4H_6.

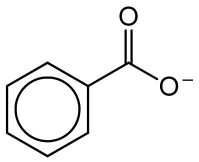
From which pair of alcohols was this hydrocarbon obtained?

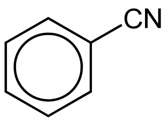
A $CH_3CH_2CH_2CH_2OH$ and $CH_3CH_2C(OH)_2CH_3$

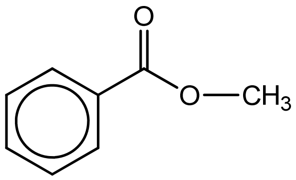
B $CH_3CH_2CH_2CH_2OH$ and $HOCH_2CH_2CH_2CH_2OH$

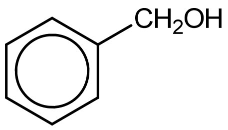
C $CH_3CH(OH)CH(OH)CH_3$ and $HOCH_2CH_2CH_2CH_2OH$

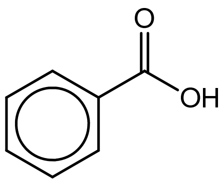
D $CH_3CH_2CH_2CH_2OH$ and $CH_3CH_2CH_2CH(OH)_2$

24. Which of the following reactions would produce  ?

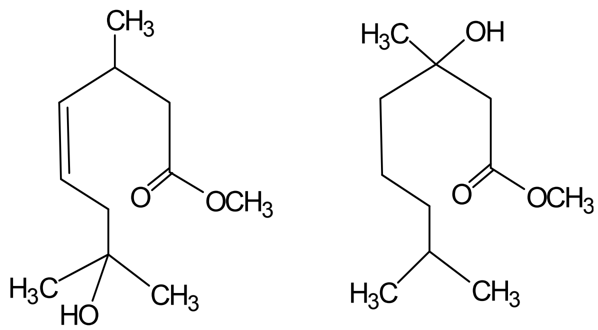
A  + hot H_2SO_4(aq)

B  + hot acidified $KMnO_4$(aq)

C  + hot acidified $K_2Cr_2O_7$(aq), distil

D  + K(s)

25. Which of the following is a suitable chemical test to distinguish the two compounds below?



A Cold dilute alkaline $KMnO_4$

B Hot dilute acidified $KMnO_4$

C Warm alkaline I_2

D HCN, trace amount of KCN

SECTION B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26. Concentrated sulfuric acid behaves as a strong acid when it reacts with water.

$$H_2SO_4(l) + aq \rightarrow H^+(aq) + HSO_4^-(aq)$$

The HSO_4^- ion formed behaves as a weak acid.

$$HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$$

Which statements are true for 1.0 mol dm^{-3} sulfuric acid?

1 $[H^+(aq)]$ is high

2 $[SO_4^{2-}(aq)]$ is high

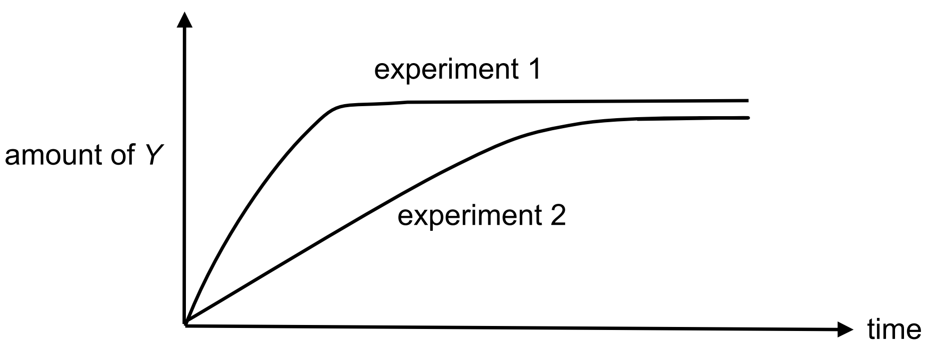
3 $[HSO_4^-(aq)] = [SO_4^{2-}(aq)]$

27. The stoichiometry of a catalysed reaction is shown by the equation below.

$$W(g) + X(g) \rightleftharpoons Y(g) + Z(g)$$

Two experiments were carried out in which the production of *Y* was measured against time.

The results are shown in the diagram below.



Which changes in the conditions from experiment 1 to experiment 2 might explain the results shown?

1 Less of *W* was used.

2 A different catalyst was used.

3 Product *Z* was continuously removed from the reaction vessel.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

28. For the hydrolysis of methyl ethanoate in aqueous hydrochloric acid, the reaction is found to be first order reaction with respect to both methyl ethanoate and hydrochloric acid.

Experiment No.	Initial concentration / mol dm^{-3}		**$t_{1/2}$ for H^+/ s**
	Methyl ethanoate	Hydrochloric acid	
1	0.10	0.20	16
2	0.05	0.10	x
3	0.10	0.40	y

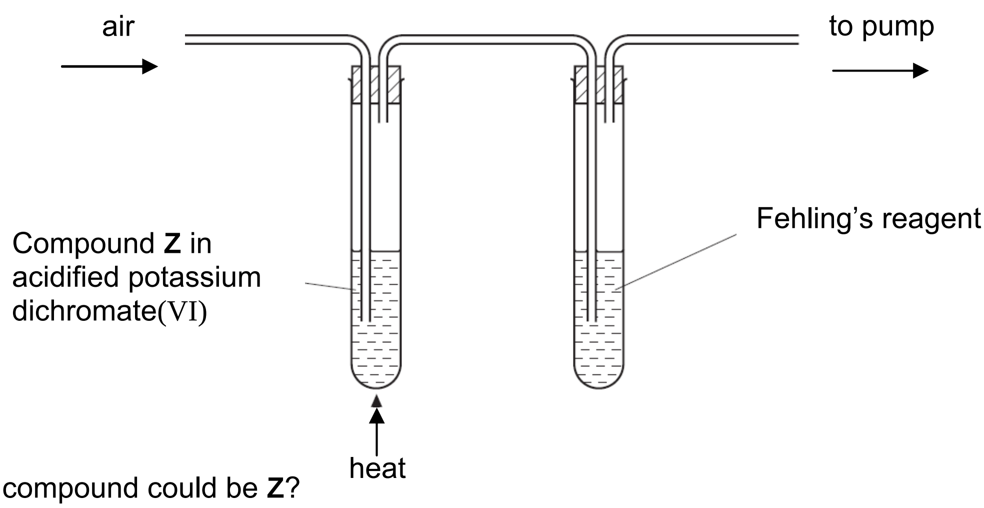
Which of the following are **not** true?

1 If the half-life of H^+ in experiment 1 is 16.0 s, then the half-life of H^+ in experiment 2 is 64.0 s.

2 If the half-life of H^+ in experiment 1 is 16.0 s, then the half-life of H^+ in experiment 3 is also 16.0 s.

3 The units for the rate constant, k , is mol^{-1} dm^3 s^{-1}.

29. When the apparatus below was used with compound **Z**, a brick-red precipitate formed in the right hand-tube.



Which compound could be **Z**?

1 $CH_3CH_2CH_2OH$

2 $CH_3CH(OH)CH_3$

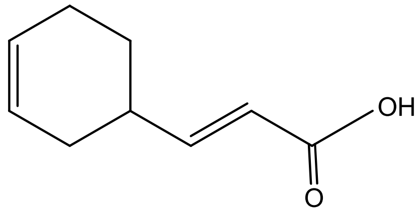
3 $C_6H_5CH_2OH$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

30. A compound **X** has the following structural formula.



From this, it can be deduced that compound **X** will **not** react with

1 $[Ag(NH_3)_2]^+$, heat

2 $NaBH_4$

3 $Na_2CO_3(aq)$

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